

Robust Pricing for Cost-Intensive Freemium Services

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This paper studies the optimal design of a two-tier freemium scheme when user demand is uncertain, usage is endogenously chosen, and the firm faces per-unit serving costs. Traditional freemium products such as mobile games have near-zero marginal costs of serving additional usage, with costs largely driven by fixed investments in content development and customer acquisition. In contrast, many recent freemium products that rely on foundational artificial intelligence models, such as large language models (LLMs) and large vision models (LVMs), incur substantial per-unit serving costs. Our framework is motivated by these emerging products and examines how freemium pricing should be designed in such environments.

We consider a continuum of users indexed by a one-dimensional type that captures their marginal value of usage. The firm offers a free plan with a usage cap at zero price and a premium plan with unlimited usage at a subscription fee. Because services based on foundational models are rapidly evolving and often lack stable historical data, point estimates of the user type distribution may be unreliable. To address this challenge, we adopt a distributionally robust approach. The firm does not know the full distribution of user types but is assumed to know certain moments of the distribution. We evaluate any freemium mechanism by its worst-case expected profit over all distributions consistent with these moment constraints.

We characterize the structure of the optimal freemium design and show that any optimal mechanism must satisfy a *collinearity condition*: the profit from free-tier users, the profit from the highest-type premium users, and the boundary between outside-option and free-plan users must be on the same line on the lower convex envelope of the profit function. We also derive comparative statics. The optimal free-plan usage cap decreases with serving costs and increases with ancillary revenue and the strength of the outside option, while the optimal premium price moves in the opposite direction. These results yield testable predictions: as inference costs for foundational models decline, free tiers should expand; platforms with stronger data complementarities should offer more generous free access; and increased competitive pressure should lead to lower premium prices.

1 Introduction

Large language model (LLM)-based services have rapidly become a standard input into consumer and enterprise workflows, with providers offering access through usage-based interfaces (queries/tokens) and tiered subscription plans. A common commercial pattern is *freemium*: a free tier that seeds adoption, paired with a premium tier that monetizes high-valuation users through a subscription. ChatGPT offers free access with message limits, Claude provides a free tier with daily usage caps, and Gemini restricts free users to a quota of queries—each balancing the imperative to attract users against the cost of serving them. In contrast to many digital settings—such as mobile gaming, streaming media, and software-as-a-service—where the marginal cost of serving an additional user session is close to zero and costs are dominated by fixed investments in content development and customer acquisition [Deng et al., 2023, Mai and Hu, 2023, Meng et al., 2021], LLM services feature substantial per-unit serving costs: each additional query and token consumes compute and incurs significant inference expense. Consequently, plan design shapes not only participation but also the equilibrium usage, directly affecting costs and profitability.

This paper studies the optimal design of a two-tier freemium scheme when (i) usage is endogenously chosen by users, (ii) the firm faces per-unit serving costs, and (iii) demand is uncertain in a way that is especially relevant for nascent services such as LLM offerings. We model a continuum of users indexed by a one-dimensional type θ capturing the marginal value of usage. The firm offers a free plan with a cap L (e.g., a token or message limit) at zero price and a premium plan with unlimited usage at a subscription price p . Given (p, L) , users choose both a plan and a usage level. The firm earns revenue from user activity (e.g., the value of user data or user feedback for model improvement) and incurs a per-unit cost of usage (e.g., inference cost). As a consequence, relaxing the free cap can increase adoption while simultaneously increasing costly usage; conversely, tightening the cap can save on costs but may push users away from the service or toward the premium tier.

Our approach captures a second feature of the LLM context: substantial uncertainty about the distribution of willingness-to-pay and usage. Early-stage or rapidly evolving services often lack stable historical data, making point estimates of the type distribution unreliable. To capture this, we adopt a distributionally robust objective: the firm does not know the full distribution of types, but it knows a moment (the mean type m) and evaluates a mechanism by its worst-case expected profit over all distributions consistent with that moment [following Carrasco et al., 2018]. This approach yields designs that are informative even when the demand environment is changing, and only coarse statistical information is available.

Contributions and main results. Our first contribution is to develop a tractable model of freemium pricing with *positive marginal serving costs*—a feature understudied in prior work. Existing freemium models [e.g., Cai and Spulber, 2023, Corrao et al., 2023, Sato, 2019] assume that marginal serving costs are zero or negligible, so that expanding the free tier costs only foregone premium revenue. This assumption is appropriate for mobile gaming and other traditional digital goods, where the primary costs are fixed—game development, server infrastructure, and customer acquisition (with per-user acquisition costs often exceeding \$20)—while the marginal cost of an additional gameplay session is essentially zero [Mai and Hu, 2023, Meng et al., 2021, Sheng et al., 2022]. In our setting, each unit of usage incurs a direct cost $\kappa > 0$, adding a new dimension to the tradeoff: a more generous free cap attracts users but also raises the firm’s cost base. This cost structure is characteristic of LLM inference, cloud computing APIs, and other cost-intensive services where resource consumption scales with usage.

Our second contribution is to characterize the structure of the optimal freemium mechanism. We show that any optimal design must satisfy a *collinearity condition*: the profit from free-tier users,

the profit from the highest-type premium users, and the boundary between outside-option and free-plan users must be aligned on the lower convex envelope of the profit function. The collinearity condition has a natural interpretation: the ratio of free-tier profit to premium-tier profit must exactly equal the ratio of the free segment length to the total active segment length. If free users are too profitable relative to their segment size, the adversary shifts mass away from them toward the less profitable endpoints; if premium users are too profitable, the adversary does the opposite. At the optimum, the mechanism is balanced so neither reallocation reduces worst-case profit. We show that an immediate implication is a price-cap tradeoff: along the collinearity constraint, higher caps require lower prices. Intuitively, a more generous free tier reduces profit per free user and expands the free segment; to maintain the required profit-to-segment ratio, premium profit must also fall, which requires lowering the price.

Our third contribution is to derive comparative statics that generate testable predictions for LLM and cost-intensive service markets. We numerically show that the optimal free cap L^* decreases with serving costs κ and increases with ancillary revenue r and outside option strength u_0 ; the optimal price p^* increases with κ and r , but decreases with u_0 . These results are consistent with observed industry patterns: (i) as inference costs have declined, providers have expanded free tiers while lowering premium prices; (ii) platforms that monetize user data offer more generous free access at higher premium prices; and (iii) as competition has intensified and outside options have improved, providers have expanded free tiers while lowering premium prices.

Related literature. Our paper contributes to three strands of literature: freemium and subscription pricing, nonlinear pricing with costs, and distributionally robust mechanism design.

Freemium pricing. Lee et al. [2017] examine the growth-versus-monetization tradeoff empirically. Sato [2019] shows that freemium emerges as the optimal menu in a two-sided market where platforms earn advertising revenue; the key tradeoff is between ad-supported free users and ad-free premium subscribers. Cai and Spulber [2023] study freemium when consumers differ in their opportunity cost of time, deriving conditions under which firms offer free-only, premium-only, or both tiers. A growing literature studies freemium in mobile gaming: Meng et al. [2021] analyze virtual currency monetization, Sheng et al. [2022] study incentivized actions such as rewarded ads, Deng et al. [2023] examine spillover effects across app portfolios, and Mai and Hu [2023] derive optimal subscription timing for multiplayer games. In these models, marginal serving costs are zero or negligible, so the free tier’s cost is purely the foregone conversion to premium (or, equivalently, the cost of acquiring users who never convert). Our model departs by introducing positive per-unit costs $\kappa > 0$, which creates a direct cost–benefit tradeoff in setting the free cap L .

Nonlinear pricing and screening with costs. The classical screening literature [Maskin and Riley, 1984, Mussa and Rosen, 1978] studies nonlinear pricing when serving different qualities or quantities is costly. Corrao et al. [2023] analyze pricing of digital goods whose usage generates ancillary revenue (e.g., data, network effects), showing that multi-part tariffs with zero-marginal-price tiers are optimal. Their model features revenue from usage rather than costs, and assumes known type distributions. Our setting combines endogenous usage, per-unit costs, and robust objectives—a combination that is increasingly common but not addressed in prior work.

Distributionally robust mechanism design. Bergemann and Schlag [2011] study robust monopoly pricing under minimax regret when the seller knows only that valuations lie in some interval. Carroll [2017] shows that when a multi-item seller knows only marginal distributions, selling items separately maximizes worst-case revenue. Carrasco et al. [2018] derive optimal selling mechanisms when only the first N moments of the value distribution are known, showing that randomization over posted prices can outperform deterministic pricing. Koçyiğit et al. [2020] characterize optimal auctions when the seller knows moment bounds on bidder values. Our paper

applies robust techniques to a freemium menu-design problem with endogenous usage and serving costs—a setting not covered by these papers, which focus on allocation mechanisms or posted prices without usage distortions.

Roadmap. Section 2 introduces the model, derives equilibrium usage under each plan, and presents the distributionally robust objective. Section 3 characterizes the cutoff types that partition users across plans. Section 4 derives the lower convex envelope geometry, rules out non-optimal configurations, and establishes the collinearity condition that pins down optimal design. Section 5 develops comparative statics, examining how the optimal mechanism responds to changes in model parameters. Section 6 discusses managerial implications for plan design in high-marginal-cost digital services. Section 7 concludes.

2 Model Setup

There is a continuum of users indexed by a taste parameter (or “type”) $\theta \in \Theta = [0, \bar{\theta}]$. Each user has quasi-linear utility over usage q . A user of type θ derives gross utility

$$u(\theta, q) = \theta q - \frac{1}{2}cq^2,$$

where $c > 0$ is a parameter capturing diminishing marginal utility. In addition, there is an outside option that delivers a constant utility $u_0 > 0$ and zero usage.

The firm offers a freemium mechanism consisting of two plans: (i) a *free plan*, denoted F , which allows usage up to a message/token limits cap L at zero monetary price; (ii) a *premium plan*, denoted P , which allows unrestricted usage at a fixed subscription price p . Given (p, L) , a user of type θ chooses a plan $a \in \{O, F, P\}$, where O denotes the outside option, and a usage level q (subject to the cap on the free plan) to maximize utility.

Outside option. Under the outside option, the user does not use the service, $q^O(\theta) = 0$, and receives utility $U^O(\theta) = u_0$.

Free plan. Under the free plan, usage is constrained by $q \leq L$ and the user pays no monetary price. For a given cap L and type θ , the user solves

$$\max_{q \leq L} \theta q - \frac{1}{2}cq^2.$$

The unconstrained maximizer of the objective is θ/c ; imposing the cap yields the optimal usage

$$q^F(\theta; L) = \min \left\{ \frac{\theta}{c}, L \right\}.$$

The corresponding indirect utility from the free plan is

$$U^F(\theta; L) = \theta q^F(\theta; L) - \frac{1}{2}c(q^F(\theta; L))^2.$$

Premium plan. Under the premium plan, there is no usage cap, and the user pays the fixed price p . For a given p and type θ , the user solves

$$\max_q \theta q - \frac{1}{2}cq^2 - p,$$

with unconstrained optimizer

$$q^P(\theta) = \frac{\theta}{c}.$$

The corresponding indirect utility from the premium plan is

$$U^P(\theta; p) = \theta q^P(\theta) - \frac{1}{2}c(q^P(\theta))^2 - p.$$

Given (p, L) , each user of type θ chooses

$$a^*(\theta; p, L) \in \arg \max_{a \in \{O, F, P\}} \{U^a(\theta; p, L)\},$$

and the induced usage is

$$q(\theta; p, L) = \begin{cases} 0 & \text{if } a^*(\theta; p, L) = O, \\ q^F(\theta; L) & \text{if } a^*(\theta; p, L) = F, \\ q^P(\theta) & \text{if } a^*(\theta; p, L) = P. \end{cases}$$

The firm earns ancillary revenue whenever users use the service, which can be the value of user data or user feedback for model improvement, or the option value of a future upgrade to more pricey plans. Let r denote the revenue per active user (per user who does not choose the outside option). Thus, for a user of type θ ,

$$R(\theta; p, L) = \begin{cases} 0 & \text{if } a^*(\theta; p, L) = O, \\ r & \text{if } a^*(\theta; p, L) = F, \\ p + r & \text{if } a^*(\theta; p, L) = P. \end{cases}$$

The firm incurs a per-unit usage cost κ , so that the cost for a type- θ user is

$$C(\theta; p, L) = \kappa q(\theta; p, L).$$

Hence, the firm's profit from a single user of type θ under (p, L) is

$$\pi(\theta; p, L) = R(\theta; p, L) - C(\theta; p, L).$$

Given a freemium mechanism (p, L) and a belief $\mu \in \mathcal{M}(m)$ over types, the firm's expected profit is

$$\Pi(p, L; \mu) = \int_{\Theta} \pi(\theta; p, L) d\mu(\theta).$$

The firm does not know the distribution of θ , but it knows that the cross-sectional mean of θ equals some $m \in (0, \bar{\theta})$. Formally, let $\mathcal{P}(\Theta)$ denote the set of Borel probability measures on Θ and define

$$\mathcal{M}(m) = \left\{ \mu \in \mathcal{P}(\Theta) : \int_{\Theta} \theta d\mu(\theta) = m \right\}.$$

The true distribution of types belongs to $\mathcal{M}(m)$ but is otherwise unrestricted. The firm is ambiguity averse about the distribution of types and thus evaluates mechanisms using the worst-case expected profit over all distributions consistent with the known mean:

$$\underline{\Pi}(p, L; m) = \inf_{\mu \in \mathcal{M}(m)} \Pi(p, L; \mu) = \inf_{\mu \in \mathcal{M}(m)} \int_{\Theta} \pi(\theta; p, L) d\mu(\theta).$$

The primary objective is to investigate the firm's choice of (p, L) to optimize this robust performance criterion.

We first show that this robust maximization problem hinges on the function $\pi(\theta; p, L)$. Fix a freemium mechanism (p, L) , define the *lower convex envelope* (LCE) of π on Θ by

$$\pi^{\text{LCE}}(\theta; p, L) = \sup \left\{ \varphi(\theta) : \varphi \text{ is convex on } \Theta \text{ and } \varphi(\tilde{\theta}) \leq \pi(\tilde{\theta}; p, L) \text{ for all } \tilde{\theta} \in \Theta \right\}.$$

By construction, $\pi^{\text{LCE}}(\cdot; p, L)$ is convex, $\pi^{\text{LCE}}(\theta; p, L) \leq \pi(\theta; p, L)$ for all $\theta \in \Theta$, and π^{LCE} is the largest convex function with this property.

The following lemma shows that the robust objective depends on (p, L) only through the lower convex envelope of π .

LEMMA 2.1 (REDUCTION TO THE LOWER CONVEX ENVELOPE). *Fix (p, L) and suppose that $\pi(\cdot; p, L)$ is bounded and Borel-measurable on Θ . Then, for every $m \in [0, \bar{\theta}]$,*

$$\underline{\Pi}(p, L; m) = \inf_{\mu \in \mathcal{M}(m)} \int_{\Theta} \pi(\theta; p, L) d\mu(\theta) = \pi^{\text{LCE}}(m; p, L).$$

Moreover, there exists a probability measure $\mu^ \in \mathcal{M}(m)$ that attains the infimum with support of cardinality at most two. In particular, there exist $\theta_1, \theta_2 \in \Theta$ with $\theta_1 \leq m \leq \theta_2$ such that*

$$\pi^{\text{LCE}}(m) = \lambda \pi(\theta_1; p, L) + (1 - \lambda) \pi(\theta_2; p, L), \quad \lambda \equiv \frac{\theta_2 - m}{\theta_2 - \theta_1},$$

and μ^ places mass λ on θ_1 and mass $1 - \lambda$ on θ_2 .*

Lemma 2.1 is a standard result in the theory of moment problems: when the only distributional constraint is the mean, there exists an extremal (worst-case) distribution with at most two support points, and the minimum expected value of any function equals its greatest convex minorant evaluated at the mean [see, e.g., Carrasco et al., 2018, Lemma 1]. The lemma establishes existence but not uniqueness; as we will see, in our setting, multiple two-point distributions may achieve the infimum. The firm knows only the mean type m and faces an adversary who selects a worst-case distribution; the lemma reduces this infinite-dimensional problem to evaluating the lower convex envelope at a single point.

Economically, Lemma 2.1 tells us that the firm should design (p, L) as if facing the most adversarial market composition: a polarized mixture of low-value and high-value users with limited middle ground. The adversary can always find such a two-point distribution that minimizes profit while matching the known mean. This worst-case perspective is particularly relevant for nascent LLM services, where the true distribution of willingness-to-pay is highly uncertain.

Lemma 2.1 implies that, for each (p, L) and each mean m , the worst-case expected profit depends only on the value of the lower convex envelope of $\pi(\cdot; p, L)$ at m . We now turn to the characterization of $\pi(\cdot; p, L)$ and $\pi^{\text{LCE}}(\cdot; p, L)$.

Before proceeding, we note that our analysis takes the two-tier freemium structure as given. This modeling choice is motivated by the prevalence of this structure in practice: ChatGPT, Claude, Gemini, and virtually all major LLM services offer exactly a free tier with usage limits and a premium tier with expanded access. Moreover, under distributional robustness, simpler mechanisms tend to perform well [Carroll, 2017], suggesting that the two-tier structure may be near-optimal even among richer mechanism classes. We leave the formal characterization of when freemium is optimal to future work.

3 Plan Choice and Cutoff Types

Understanding how users sort across plans is essential for characterizing optimal design. The firm's worst-case profit depends critically on which types choose the outside option, which use the free tier, and which upgrade to premium. Each segment contributes differently to profit, and the adversary will exploit any imbalance. We now derive the cutoff types that determine this partition and characterize the resulting profit function.

Fix (p, L) , using $q^F(\theta; L) = \min\{\theta/c, L\}$ and $q^P(\theta) = \theta/c$, we can obtain

$$U^F(\theta; L) = \begin{cases} 1/2c\theta^2 & \text{if } \theta \leq cL, \\ \theta L - 1/2cL^2 & \text{if } \theta > cL, \end{cases} \quad U^P(\theta; p) = \frac{1}{2c}\theta^2 - p.$$

We now characterize the cutoff types that separate customers who optimally choose the outside option, the free plan, and the premium plan. We first derive the cutoff types for users to choose among these options.

The outside option–free plan cutoff. Let θ_{OF} denote a type that is indifferent between the outside option and the free plan,

$$U^O(\theta_{OF}) = U^F(\theta_{OF}; L).$$

There are two cases, depending on whether the free-plan cap binds at the cutoff. First, unconstrained usage at the cutoff: If $\theta_{OF} < cL$, then $q^F(\theta_{OF}; L) = \theta_{OF}/c$ and $u_0 = U^F(\theta_{OF}; L) = \theta_{OF}^2/(2c)$, so that $\theta_{OF}(L) = \sqrt{2cu_0}$. Consistency of this case requires $\theta_{OF} < cL$, i.e., $L > \sqrt{2u_0}/c$. Second, binding cap at the cutoff: If instead $\theta_{OF} \geq cL$, then $q^F(\theta_{OF}; L) = L$ and $u_0 = U^F(\theta_{OF}; L) = \theta_{OF}L - cL^2/2$, so that $\theta_{OF}(L) = u_0/L + cL/2$. Consistency now requires $\theta_{OF} \geq cL$, i.e., $L \leq \sqrt{2u_0}/c$. Combining the two cases, we can write θ_{OF} as a function of L

$$\theta_{OF}(L) = \begin{cases} \frac{u_0}{L} + \frac{1}{2}cL & \text{if } L \leq \sqrt{2u_0}/c, \\ \sqrt{2cu_0} & \text{if } L > \sqrt{2u_0}/c. \end{cases}$$

Note that $\theta_{OF}(L)$ does not depend on p . When $L < \sqrt{2u_0}/c$, we have

$$\frac{\partial \theta_{OF}(L)}{\partial L} = -\frac{u_0}{L^2} + \frac{1}{2}c < 0.$$

Thus, for small caps, an increase in L makes the free plan more attractive and shifts the outside option–free boundary to lower types; once L is large enough that the cap does not bind at the cutoff, the boundary becomes locally insensitive to further increases in L .

The free–premium plan cutoff. Let θ_{FP} denote a type that is indifferent between the free and premium plans,

$$U^F(\theta_{FP}; L) = U^P(\theta_{FP}; p).$$

Then we can write θ_{FP} as a function of p and L

$$\theta_{FP}(p, L) = cL + \sqrt{2cp}.$$

Note that this free–premium plan boundary is strictly increasing in both p and L . Intuitively, a higher subscription price or a more generous cap makes the premium plan less attractive relative to the free plan, so that only higher types are willing to pay for the premium plan.

Segments of types. It is without loss of generality to restrict attention to mechanisms (p, L) such that $\theta_{OF}(L) \leq \theta_{FP}(p, L)$. When the inequality is strict, types between θ_{OF} and θ_{FP} strictly prefer the free plan; when $\theta_{OF} = \theta_{FP}$, no type strictly prefers free, and all users choose either the outside option or premium. Any configuration without free users can be equivalently represented by a boundary mechanism where all three options are equally attractive at the cutoff type (see Appendix A.1 for the construction). This restriction therefore nests all economically distinct cases, including the knife-edge where no type chooses free.

Furthermore, we henceforth focus on mechanisms (p, L) for which a non-degenerate set of users choose the premium plan. This corresponds to assuming that the two cutoffs lie strictly inside the support of types and are ordered as

$$\theta_{OF}(L) \leq \theta_{FP}(p, L) < \bar{\theta}.$$

Under this condition, the population is partitioned into three segments under mechanism (p, L) :

- (1) *Outside-option users:* $\theta \in [0, \theta_{OF}]$, for whom $a^*(\theta; p, L) = O$.

- (2) *Free-plan users*: $\theta \in (\theta_{OF}, \theta_{FP}]$, for whom $a^*(\theta; p, L) = F$. Note that the case in which no type chooses the free plan is represented by $\theta_{OF} = \theta_{FP}$.
- (3) *Premium-plan users*: $\theta \in (\theta_{FP}, \bar{\theta}]$, for whom $a^*(\theta; p, L) = P$.

These cutoffs have intuitive comparative statics that illuminate how the mechanism shapes market segmentation. Raising the cap L shifts θ_{OF} to the left (encouraging adoption by making the free plan more attractive relative to the outside option) and shifts θ_{FP} to the right (encouraging users to stay on the free plan rather than upgrading to premium). Raising the price p leaves θ_{OF} unchanged but shifts θ_{FP} to the right, as the premium plan becomes less attractive. The firm thus faces the tradeoff that a generous free tier expands adoption but cannibalizes premium subscriptions, while a restrictive free tier preserves premium revenue but risks losing users to the outside option.

At the cutoff types themselves, preferences across plans are aligned by our tie-breaking rule: when indifferent, the user chooses the plan that is worst for the firm. This is consistent with our consideration of worst-case expected profit and will be convenient when we characterize the lower convex envelope in the next section.

The profit function. Given these segments, we can now characterize how the firm's profit varies across user types. Given the cutoffs θ_{OF} and θ_{FP} , we can write the firm's profit from a type- θ user in the following two cases: (i) when the free-plan cap is not binding for the type θ_{OF} , i.e., $L > \sqrt{2u_0}/c$, we can write

$$\pi(\theta; p, L) = \begin{cases} 0 & \text{if } \theta \leq \theta_{OF}, \\ r - \kappa\theta/c & \text{if } \sqrt{2cu_0} < \theta \leq cL, \\ r - \kappa L & \text{if } cL < \theta \leq \theta_{FP}, \\ p + r - \kappa\theta/c & \text{if } \theta > \theta_{FP}; \end{cases}$$

(ii) when the free-plan cap binds for the type θ_{OF} , i.e., $L \leq \sqrt{2u_0}/c$, we can write

$$\pi(\theta; p, L) = \begin{cases} 0 & \text{if } \theta \leq \theta_{OF}, \\ r - \kappa L & \text{if } \theta_{OF} < \theta \leq \theta_{FP}, \\ p + r - \kappa\theta/c & \text{if } \theta > \theta_{FP}. \end{cases}$$

Thus, $\pi(\cdot; p, L)$ is piecewise linear in θ , with jumps or kinks at θ_{OF} , cL , and θ_{FP} , and the locations of these kinks respond to (p, L) according to the comparative statics derived above. In particular, increasing p or L shifts the free-premium boundary $\theta_{FP}(p, L)$ upward, while increasing L (for small caps) shifts the outside option-free boundary $\theta_{OF}(L)$ downward.

Note that the profit function in case (i) is more complex due to the presence of unconstrained free users. A natural question is whether optimal mechanisms ever induce such unconstrained usage. The following lemma shows that the answer is no: optimal mechanisms always bind the cap for all free users.

LEMMA 3.1 (BINDING-CAP OPTIMALITY). *Any optimal mechanism satisfies $L^* \leq \sqrt{2u_0}/c$, ensuring that all free-plan users are constrained by the usage cap.*

PROOF. See Appendix A.2. □

The intuition is that allowing unconstrained free usage is wasteful: when some free users consume below the cap, the firm can simultaneously tighten the cap and raise the premium price, reducing costs on free users while extracting more surplus from premium users. This joint adjustment improves profit across all types without changing the boundaries between segments. Consequently, we may restrict attention to mechanisms for which the profit function takes the simpler binding-cap form in case (ii).

We discuss several assumptions before we turn to the LCE. We impose $r > \kappa\sqrt{2u_0/c}$, which implies $r - \kappa L > 0$ for all $L \leq \sqrt{2u_0/c}$. Hence, every free-plan user generates a strictly positive net profit. Although the direct ancillary revenue per user (e.g., advertising impressions) may be modest, we interpret r broadly to include several components that are particularly valuable for LLM providers: (i) the expected present value of training data and user feedback, which improves model quality and reduces future development costs; (ii) the option value of future conversion to premium plans; and (iii) demonstration and network effects that enhance the firm's market position. For nascent AI services, where each user interaction contributes to model improvement, the aggregate value of these components can substantially exceed the per-user serving cost κL at moderate usage caps.

When ancillary revenue is low (e.g., r close to zero), free-plan users may generate negative profits: the serving cost κL exceeds the revenue r . In this regime, offering a free tier becomes purely a loss-leader strategy, and the worst-case profit can be negative whenever a free plan is offered. This leads to a relatively uninteresting case where the firm abandons the free tier entirely. Our assumption $r > \kappa\sqrt{2u_0/c}$ allows us to focus on the more interesting case where the freemium model is viable—free users contribute positively to profit, making the tradeoff between adoption and monetization economically meaningful.

In addition, we assume that the parameters are such that there exists at least one mechanism (p, L) yielding strictly positive worst-case profit, i.e., $\pi^{\text{LCE}}(m; p, L) > 0$ for the given mean m . This rules out degenerate environments in which all mechanisms generate zero robust profit. It also implies that an optimal mechanism must satisfy $\theta_{OF}(L) < m$, because otherwise $\pi^{\text{LCE}}(m; p, L) = 0$.

4 Characterization of the Lower Convex Envelope

With the profit function characterized, we now turn to the geometry that determines worst-case profit. The key insight is that the piecewise linear structure of $\pi(\cdot; p, L)$ limits the possible shapes of its lower convex envelope. Because π is affine on each of the three intervals $[0, \theta_{OF}]$, $(\theta_{OF}, \theta_{FP}]$, and $(\theta_{FP}, \bar{\theta}]$, the LCE can only “touch” π at the endpoints of these intervals. Understanding which configurations of these contact points are sustainable under adversarial demand—and which are not—reveals the geometric structure that optimal mechanisms must satisfy.

We now characterize the lower convex envelope $\pi^{\text{LCE}}(\cdot; p, L)$ under an optimal mechanism (p, L) .

First, only four points, 0, θ_{OF} , θ_{FP} , and $\bar{\theta}$ can determine the LCE. The profit function $\pi(\cdot; p, L)$ is affine on each of the three intervals $[0, \theta_{OF}]$, $(\theta_{OF}, \theta_{FP}]$, and $(\theta_{FP}, \bar{\theta}]$, with slopes 0, 0, and $-\kappa/c$, respectively. Any convex minorant fits each affine piece with a line and therefore can only “touch” π at the endpoints of these affine pieces. No interior point can become a vertex of the envelope. Hence $\pi^{\text{LCE}}(\cdot; p, L)$ can only have vertices chosen among the four points above.

We now discuss which subsets of these four points can actually lie on the lower convex envelope. Let us write, for brevity,

$$\pi_F \equiv r - \kappa L > 0, \quad \pi_H \equiv \pi(\bar{\theta}; p, L) = p + r - \kappa \frac{\bar{\theta}}{c}.$$

Since π jumps from 0 at θ_{OF} to π_F on $(\theta_{OF}, \theta_{FP}]$, and then falls linearly to π_H at $\bar{\theta}$, the relative position of π_F and π_H yields four qualitatively distinct cases (see Figure 1):

- (a) High types unprofitable ($\pi_H < 0$): The chord from $(0, 0)$ to $(\bar{\theta}, \pi_H)$ lies below $\pi(\theta; p, L)$ for all θ . Then the LCE uses only 0 and $\bar{\theta}$ as vertices.
- (b) Moderately profitable but low π_H : The chord from $(\theta_{OF}, 0)$ to $(\bar{\theta}, \pi_H)$ lies strictly below the free-plan plateau $r - \kappa L$. Then the LCE has vertices at 0, θ_{OF} , and $\bar{\theta}$, but the intermediate

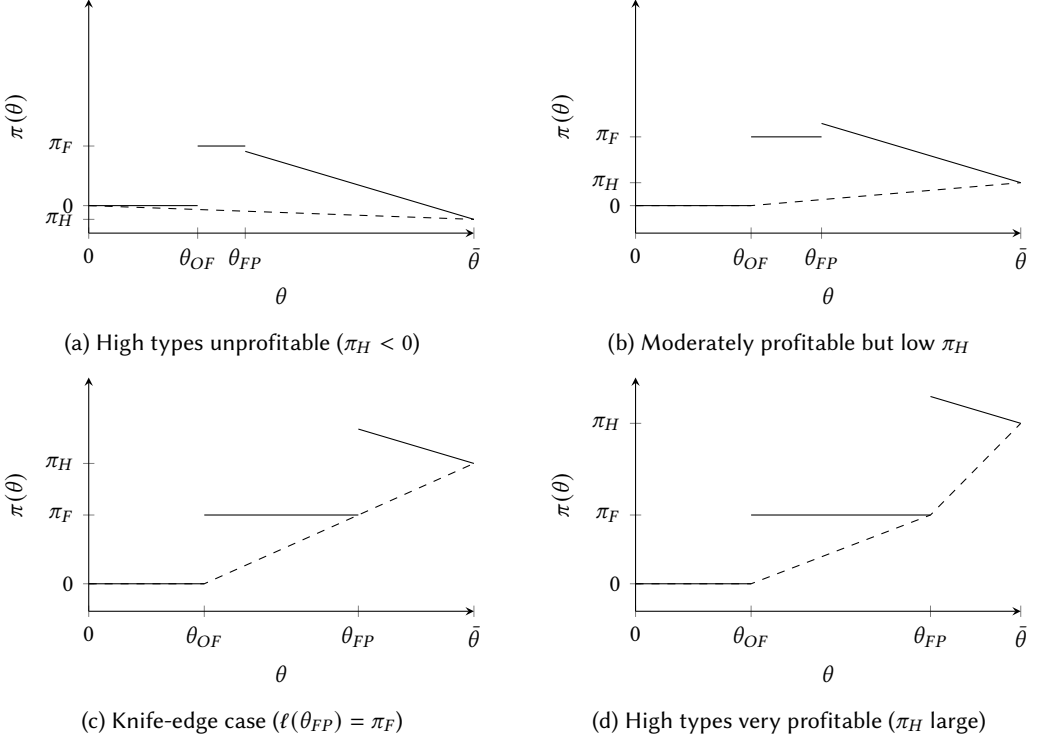


Fig. 1. Illustrations of the four lower-convex-envelope cases for $\theta \in [0, 5]$ under $c = 1$, $u_0 = 1$, $\kappa = 0.5$, $r = 2$.

Notes. The solid curve is the pointwise profit function $\pi(\theta; p, L)$ induced by the freemium mechanism: it is 0 for $\theta \leq \theta_{OF}(L)$, equals $\pi_F = r - \kappa L$ on $(\theta_{OF}(L), \theta_{FP}(p, L))$, and is affine on $[\theta_{FP}(p, L), \bar{\theta}]$. The dashed curve is the lower convex envelope $\pi^{\text{LCE}}(\cdot; p, L)$, i.e., the largest convex function on Θ dominated by $\pi(\cdot; p, L)$. As in the text, only the four points 0 , θ_{OF} , θ_{FP} , and $\bar{\theta}$ can be vertices of the LCE; each panel corresponds to one of the four qualitatively distinct configurations.

point θ_{FP} is not used. Then, the LCE is given by

$$\pi^{\text{LCE}}(\theta; p, L) = \begin{cases} 0 & \text{for } \theta \in [0, \theta_{OF}], \\ \text{chord between } (\theta_{OF}, 0) \text{ and } (\bar{\theta}, \pi_H) & \text{for } \theta \in [\theta_{OF}, \bar{\theta}]. \end{cases}$$

- (c) Knife-edge case. The chord from $(\theta_{OF}, 0)$ to $(\bar{\theta}, \pi_H)$ passes exactly through (θ_{FP}, π_F) . Then θ_{OF} , θ_{FP} , and $\bar{\theta}$ lie on a single supporting line, and all four candidate points appear as vertices (with three collinear on the upper branch). The resulting LCE is identical to the previous case in functional form.
- (d) High types sufficiently profitable (π_H large): The chord from $(\theta_{OF}, 0)$ to $(\bar{\theta}, \pi_H)$ lies above the plateau π_F on some subinterval of $[\theta_{OF}, \theta_{FP}]$. In that case, the convex envelope must “touch” π again before reaching $\bar{\theta}$ at θ_{FP} . Then, the LCE is given by

$$\pi^{\text{LCE}}(\theta; p, L) = \begin{cases} 0 & \text{for } \theta \in [0, \theta_{OF}], \\ \text{chord between } (\theta_{OF}, 0) \text{ and } (\theta_{FP}, \pi_F) & \text{for } \theta \in [\theta_{OF}, \theta_{FP}], \\ \text{chord between } (\theta_{FP}, \pi_F) \text{ and } (\bar{\theta}, \pi_H) & \text{for } \theta \in [\theta_{FP}, \bar{\theta}]. \end{cases}$$

Before stating our main result, we develop intuition for why only the knife-edge case (c) is consistent with optimality.

In case (a), the premium price is too low: the highest types consume the most and generate negative profit. The firm can always raise p , increasing profit on premium users without affecting other segments. In case (b), the premium price is still too low relative to the free tier's profitability: the adversary can concentrate mass on free users and the highest types, exploiting the fact that the chord lies below the free plateau. Raising p steepens the chord and increases worst-case profit. In case (d), the premium price is too high: the premium segment is disproportionately profitable relative to the free segment. The adversary exploits this imbalance by shifting mass toward lower types. The firm can improve by either lowering p (if the mean is below θ_{FP}) or lowering L (if the mean is above θ_{FP}).

Only case (c)—the knife-edge collinearity—achieves balance: no local perturbation of p or L can unambiguously increase worst-case profit. The three key points are perfectly aligned, leaving the adversary no room for exploitation.

The following proposition establishes that any optimal mechanism must induce the knife-edge third case.

PROPOSITION 4.1 (COLLINEARITY CHARACTERIZATION). *Let (p^*, L^*) be an optimal freemium mechanism. Then the mechanism satisfies the following collinearity condition: the points $(\theta_{OF}(L^*), 0)$, $(\theta_{FP}(p^*, L^*), \pi_F(L^*))$, and $(\bar{\theta}, \pi_H(p^*))$ are collinear. Equivalently,*

$$\frac{\pi_F(L^*)}{\theta_{FP}(p^*, L^*) - \theta_{OF}(L^*)} = \frac{\pi_H(p^*)}{\bar{\theta} - \theta_{OF}(L^*)}.$$

PROOF SKETCH. We show that only the knife-edge third case (Figure 1c) is consistent with optimality; the other three cases admit local improvements. The full argument is in Appendix A.3.

Case (a): If $\pi_H < 0$, increasing p raises π_H without affecting other segments, strictly improving the LCE everywhere.

Case (b): If the chord from $(\theta_{OF}, 0)$ to $(\bar{\theta}, \pi_H)$ lies below π_F , raising p steepens the chord and increases the LCE at the mean, since an optimal mechanism requires $m > \theta_{OF}$.

Case (d): If the chord lies above π_F , the premium segment is disproportionately profitable. If $m < \theta_{FP}$, lowering p compresses the free segment and raises the LCE at m ; if $m \geq \theta_{FP}$, lowering L raises π_F and shrinks the free segment, again improving the LCE at m .

Only case (c) admits no improving perturbation: the three key points are collinear, leaving no room for the adversary to exploit an imbalance. $\square$

An immediate consequence of the collinearity characterization is that optimal mechanisms must feature a non-trivial free tier—the firm cannot optimally exclude free users entirely.

COROLLARY 4.2 (NON-DEGENERACY OF FREE TIER). *Under the collinearity condition, the free tier is non-degenerate: $\theta_{OF}(L^*) < \theta_{FP}(p^*, L^*)$.*

PROOF. When $\theta_{OF} = \theta_{FP}$, no type strictly prefers the free plan—users either choose the outside option or jump directly to premium. This corresponds to Case (b) in Figure 1, which was ruled out in the proof of Proposition 4.1. $\square$

Economic interpretation. The collinearity condition can be rewritten as

$$\frac{\pi_F}{\pi_H} = \frac{\theta_{FP} - \theta_{OF}}{\bar{\theta} - \theta_{OF}},$$

which says that the ratio of free-tier profit to premium-tier profit exactly equals the ratio of the free segment length to the total active segment length. This balance ensures the mechanism is robust

against adversarial reallocation of probability mass. If free users were “too profitable” relative to their segment size (i.e., $\pi_F/\pi_H > (\theta_{FP} - \theta_{OF})/(\bar{\theta} - \theta_{OF})$), the adversary would shift mass away from them toward the less profitable endpoints. Conversely, if premium users were too profitable, the adversary would shift mass away from the high end. At the optimum, neither reallocation can reduce worst-case profit: the adversary is indifferent among all worst-case distributions with the same mean, and the firm’s robust payoff depends smoothly on the market’s composition.

The collinearity condition links the two design instruments, p and L , through the balance requirement. This linkage implies that the firm cannot independently optimize the free cap and the premium price—adjusting one necessitates adjusting the other. The following corollary formalizes this interdependence.

COROLLARY 4.3 (PRICE-CAP TRADEOFF). *Along the collinearity constraint, $dp/dL < 0$: a more generous free cap requires a lower premium price.*

PROOF. Let $A = \bar{\theta} - \theta_{OF}$ and $B = \theta_{FP} - \theta_{OF}$. The collinearity constraint is $\pi_F A = \pi_H B$. Differentiating both sides with respect to L :

$$\frac{d\pi_F}{dL} A + \pi_F \frac{dA}{dL} = \frac{d\pi_H}{dL} B + \pi_H \frac{dB}{dL}.$$

Using $d\pi_F/dL = -\kappa$, $d\pi_H/dL = dp/dL$, $dA/dL = -d\theta_{OF}/dL$, and $dB/dL = c + (dp/dL)\sqrt{c/(2p)} - d\theta_{OF}/dL$, we obtain after rearranging:

$$\frac{dp}{dL} = \frac{-\kappa A + (\pi_H - \pi_F)d\theta_{OF}/dL - \pi_H c}{B + \pi_H \sqrt{c/(2p)}}.$$

The denominator is positive. For the numerator: (i) $-\kappa A < 0$; (ii) collinearity implies $\pi_F/\pi_H = B/A < 1$, so $\pi_H > \pi_F$, and since $d\theta_{OF}/dL < 0$ for $L < \sqrt{2u_0/c}$, we have $(\pi_H - \pi_F)(d\theta_{OF}/dL) < 0$; (iii) $-\pi_H c < 0$. All three terms are negative, so $dp/dL < 0$. $\square$

Interpretation. The price-cap tradeoff reflects an important tension in freemium design. When L increases: (i) θ_{OF} shifts left (more types adopt); (ii) $\pi_F = r - \kappa L$ decreases (free users become less profitable); (iii) θ_{FP} shifts right (more types choose free over premium). To maintain the collinearity condition—which equates profit rates across segments—the firm must lower p , reducing π_H and the slope of the LCE. Intuitively, a generous free tier attracts users away from premium, so the premium price must fall to restore balance.

The collinearity condition in Proposition 4.1 can be rewritten in terms of model primitives as

$$(r - \kappa L)(\bar{\theta}L - u_0 - \frac{\epsilon}{2}L^2) = \left(p + r - \kappa \frac{\bar{\theta}}{c}\right) \left(\frac{\epsilon}{2}L^2 + L\sqrt{2cp} - u_0\right).$$

Proposition 4.1 reduces the firm’s robust optimization problem to a tractable constrained program.

PROPOSITION 4.4 (REDUCED OPTIMIZATION PROBLEM). *Under the collinearity condition, the worst-case profit is*

$$\underline{\Pi}(p, L; m) = \frac{m - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p),$$

and the firm’s optimization problem reduces to

$$\begin{aligned} \max_{p, L} \quad & \left(p + r - \kappa \frac{\bar{\theta}}{c} \right) \frac{cL^2 - 2mL + 2u_0}{cL^2 - 2\bar{\theta}L + 2u_0}, \\ \text{s.t.} \quad & (r - \kappa L)(\bar{\theta}L - u_0 - \frac{\epsilon}{2}L^2) = \left(p + r - \kappa \frac{\bar{\theta}}{c}\right) \left(\frac{\epsilon}{2}L^2 + L\sqrt{2cp} - u_0\right), \end{aligned}$$

$$\frac{m - \sqrt{m^2 - 2cu_0}}{c} < L \leq \sqrt{\frac{2u_0}{c}}.$$

The collinearity constraint ensures that the optimal cap L^* is strictly positive and that a non-degenerate interval of types chooses the free plan.

Proposition 4.4 transforms the robust pricing problem from an infinite-dimensional optimization over distributions to a tractable two-variable constrained program. The objective function captures how worst-case profit depends on the mechanism parameters, while the collinearity constraint encodes the balance requirement from Proposition 4.1. Although the constraint is nonlinear (due to the $\sqrt{2c\bar{p}}$ term in θ_{FP}), the problem is low-dimensional and amenable to numerical solution. The bound on L reflects Lemma 3.1 (binding-cap optimality) and the requirement that the mean m exceed $\theta_{OF}(L)$ for positive worst-case profit. In the next section, we use this reduced formulation to investigate how optimal pricing responds to changes in the underlying cost and demand parameters.

5 Comparative Statics

We now examine how the optimal mechanism (p^*, L^*) responds to changes in the key exogenous parameters. To do so, we numerically solve the constrained optimization problem in Proposition 4.4 for varying parameter values. For the baseline case, we set $c = 1$, $u_0 = 1$, $\kappa = 0.5$, $r = 2$, $\bar{\theta} = 5$, and $m = 2.5$. The baseline optimal mechanism is $(p^*, L^*) \approx (3.35, 0.87)$ with worst-case profit $\Pi^* \approx 0.76$. Figure 2 illustrates how the optimal price p^* (left axis, solid blue) and cap L^* (right axis, dashed red) respond to changes in the serving cost κ , the ancillary revenue r , and the outside option utility u_0 .

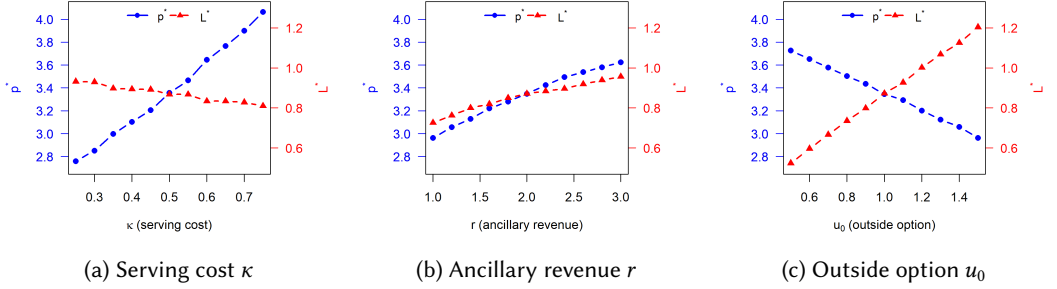


Fig. 2. Comparative statics of the optimal freemium mechanism. Each panel shows how the optimal price p^* (left axis, solid blue with circles) and usage cap L^* (right axis, dashed red with triangles) vary with one parameter, holding others at baseline values ($c = 1$, $u_0 = 1$, $\kappa = 0.5$, $r = 2$, $\bar{\theta} = 5$, $m = 2.5$). The y-axes use consistent scales across panels to facilitate comparison.

Serving cost κ (Panel a). As the per-unit serving cost κ increases, the optimal cap L^* decreases monotonically while the optimal price p^* increases. The mechanism is as follows: higher serving costs directly reduce the profit from free-tier users, $\pi_F = r - \kappa L$. When free users become more expensive to serve, the firm tightens the cap to limit its cost exposure. At the same time, higher κ reduces the profit from premium users, $\pi_H = p + r - \kappa \bar{\theta}/c$, because the highest-type users consume the most and thus impose the highest costs. To offset this reduction in π_H , the firm raises the subscription price p^* . The price-cap tradeoff established in Corollary 4.3 reinforces this pattern: as L^* falls, the collinearity condition requires p^* to rise.

This comparative static—absent from zero-marginal-cost freemium models—is the distinguishing prediction of our framework. It implies that *as LLM inference costs decline over time, providers should*

expand their free tiers. This prediction aligns with observed industry trends: as GPT-4-class models become cheaper to run, providers such as OpenAI and Anthropic have progressively increased free-tier allowances. Conversely, during periods of capacity constraints or high inference costs, we should expect tighter free-tier limits.

Ancillary revenue r (Panel b). Higher ancillary revenue leads to both a more generous free cap and a higher subscription price. Unlike the serving cost case, here both instruments move in the same direction. The intuition is that r enters both $\pi_F = r - \kappa L$ and $\pi_H = p + r - \kappa \bar{\theta}/c$ symmetrically with a coefficient of +1. When each active user generates more value—through data contributions, feedback for model improvement, or option value for future upgrades—the firm can afford to subsidize additional free usage. The direct increase in π_F makes free users more valuable, justifying a larger cap L^* . Simultaneously, the increase in π_H allows the firm to extract more surplus through premium subscriptions, supporting a higher price p^* .

This suggests that *platforms deriving substantial value from user data should offer more generous free access.* LLM providers that use user interactions for model fine-tuning or RLHF (reinforcement learning from human feedback) effectively have higher r than providers who do not monetize user data. Our model predicts that data-centric platforms will offer more generous free tiers, cross-subsidizing usage costs with the value of collected data.

Outside option u_0 (Panel c). A stronger outside option leads to a more generous cap and a lower subscription price. When users have better alternatives—such as competing LLM services or open-source models—the adoption threshold $\theta_{OF}(L) = u_0/L + cL/2$ rises, shrinking the pool of users who prefer the free plan over opting out. To restore adoption, the firm must increase L^* , which lowers θ_{OF} and makes the free plan more attractive relative to the outside option. By the price-cap tradeoff (Corollary 4.3), this increase in L^* requires a corresponding decrease in p^* . The firm thus responds on both margins: a more generous free tier to attract users at the adoption boundary, and a lower premium price to retain users who might otherwise opt out entirely.

This prediction implies that *increased competition in the LLM market will push providers toward more generous free tiers and lower subscription prices.* As the market matures and users gain access to more alternatives, the effective outside option u_0 rises. Our model predicts that this competitive pressure will benefit consumers through both expanded free access and lower premium pricing—a form of welfare transfer from producers to consumers that intensifies as the market becomes more competitive.

6 Managerial Implications

Balancing free and premium profitability. The collinearity condition provides a concrete design principle: the ratio of profit per free user to profit per premium user must equal the ratio of their respective segment sizes in the type space. This balance has immediate practical implications. If the free tier is too generous—high L leading to low $\pi_F = r - \kappa L$ —the firm over-invests in users who generate little profit relative to the type space they occupy. Conversely, if the premium price is too high, the firm under-serves the premium segment. The optimal design equates these margins, ensuring that neither segment is disproportionately profitable or unprofitable relative to its size.

For managers, this suggests a diagnostic: estimate the profit contribution of free versus premium users and compare it to the fraction of active users in each segment. Consider a concrete example: suppose free users generate \$0.50 profit per user (from data value minus serving costs) while occupying 60% of active users, and premium users generate \$2.00 profit per user while occupying 40%. The profit ratio is $0.50/2.00 = 0.25$, but the segment ratio is $0.60/1.00 = 0.60$. This imbalance—free users are less profitable per capita than their segment share would warrant—suggests the cap is too generous. Tightening L would raise π_F and shrink the free segment, restoring balance.

The price-cap tradeoff. A key constraint facing managers is that the free cap and premium price cannot be set independently. The collinearity condition implies that higher caps require lower prices: expanding the free tier reduces profit per free user and enlarges the free segment, which forces a reduction in premium profit to maintain balance. This tradeoff means that managers cannot simultaneously offer a very generous free tier and charge a high premium price—attempting to do so would create an imbalance that leaves profit on the table.

In practice, this suggests that firms should view cap and price decisions as joint rather than separate. A decision to expand the free tier (e.g., in response to competitive pressure) should be accompanied by a reassessment of premium pricing. For example, if a firm increases its free message limit from 50 to 100 messages per day, it should simultaneously evaluate whether the premium subscription price needs to decrease to maintain the collinearity balance. Conversely, if the firm wishes to raise premium prices, it should expect the optimal cap to tighten. This joint perspective prevents the common mistake of treating free-tier generosity and premium pricing as independent marketing decisions.

Responding to changing market conditions. The comparative statics provide guidance for dynamic adjustment. As inference costs decline—a trend that has characterized the LLM industry—firms should expand their free tiers while lowering premium prices. The dramatic reduction in inference costs for GPT-4-class models between 2023 and 2025 corresponds, in our model, to a decline in κ , which predicts exactly the expansion of free tiers observed in practice. Firms that invest in monetizing user data (e.g., through model fine-tuning or RLHF) effectively increase their ancillary revenue r , which justifies both a more generous free cap and a higher premium price. As competition intensifies and users’ outside options improve—as occurred with the entry of Claude, Gemini, and open-source alternatives into the LLM market—firms must respond by offering more generous free tiers and lower premium prices. Our model predicts this competitive dynamic, which benefits consumers through both expanded free access and lower subscription costs.

Profit guarantees under uncertainty. The robust design provides a profit floor: if the optimal mechanism achieves worst-case profit $\underline{\Pi}^*$, the firm is guaranteed at least this profit regardless of the true type distribution (as long as it has the assumed mean m). Realized profits under actual market conditions—which typically feature a richer distribution of types than the sparse worst-case distribution—should meet or exceed this guarantee. In our baseline numerical example, the worst-case profit is approximately 0.76 per unit of mean type. If the actual distribution is more favorable (e.g., a uniform or bell-shaped distribution rather than the adversarial two-point distribution), realized profits will exceed this floor. This profit guarantee makes robust freemium design particularly valuable for nascent services where demand uncertainty is substantial: the firm can confidently deploy the mechanism knowing that profits will be at least $\underline{\Pi}^*$, while likely performing better in practice.

When to use robust versus Bayesian pricing. The robust approach is most valuable when historical data are scarce or user behavior is evolving rapidly. As the service matures and the firm accumulates data, it may transition to Bayesian pricing with a calibrated prior. The robust design provides a principled starting point that guards against costly mistakes during the learning phase, and the profit guarantee ensures that the firm does not suffer large losses while gathering information about the market.

7 Conclusion

This paper develops a model of freemium pricing that incorporates a feature central to LLM and cost-intensive services: positive marginal serving costs. Adopting a distributionally robust objective

to address demand uncertainty, we characterize the optimal freemium mechanism through a geometric condition—collinearity on the lower convex envelope of the profit function—that reduces the robust optimization problem to a tractable two-variable program. The collinearity condition has an intuitive economic interpretation: the ratio of free-tier profit to premium-tier profit must equal the ratio of their respective segment sizes, ensuring that neither tier is disproportionately profitable relative to its share of the market. An immediate implication is a price-cap tradeoff: higher caps require lower prices, so managers must view these as joint decisions.

The central insight is that positive marginal costs substantially change freemium design. In traditional zero-cost settings, the free tier is purely a customer acquisition tool; in our setting, it is a profit center that must be balanced against the premium tier. This balance—captured by the collinearity condition—provides a concrete design principle that managers can use to diagnose and correct imbalances in their freemium offerings.

Our numerical comparative statics are consistent with observed industry patterns. As inference costs have declined, providers have expanded free tiers while lowering premium prices. Platforms that monetize user data offer more generous free access at higher premium prices. And as competition has intensified, free tiers have expanded while premium prices have fallen. The robust design provides a profit guarantee: realized profits under actual market conditions should meet or exceed the worst-case bound, making this approach particularly valuable for nascent services where demand uncertainty is substantial.

Several extensions merit future investigation. First, while our comparative statics are established numerically, deriving analytical closed-form results would yield sharper theoretical predictions and deeper insight into the mechanism. The nonlinear nature of the collinearity constraint makes this challenging but potentially tractable under specific assumptions. Second, we assume a two-tier freemium structure. Analyzing richer mechanisms—such as multiple tiers, usage-based pricing, or nonlinear tariffs—and characterizing when the simple freemium structure is optimal would strengthen the foundations of our analysis. Third, our model considers a single firm and captures competition through a utility term of the outside option. Incorporating strategic competition would enable analysis of equilibrium levels of free-tier generosity and premium pricing and could reveal whether competitive pressure leads to excessive free-tier generosity or a race to the bottom in premium pricing. Fourth, we assume a static environment. A dynamic model could capture learning about user types over time and the option value of using the free tier as an experimentation platform. Finally, empirical validation using data from LLM providers—testing whether observed pricing patterns match our theoretical predictions—would complement the analysis and provide guidance for calibrating the model parameters.

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A Technical Proofs and Derivations

This appendix contains detailed proofs and technical arguments that are summarized in the main text.

A.1 Without-Loss-of-Generality Restriction on Cutoffs

We show that restricting attention to mechanisms (p, L) with $\theta_{OF}(L) \leq \theta_{FP}(p, L)$ is without loss of generality.

Suppose that, for some subscription price p , there exists a cutoff type $\tilde{\theta}$ such that all users with $\theta \leq \tilde{\theta}$ choose the outside option, while all users with $\theta > \tilde{\theta}$ choose the premium plan, and no user chooses the free plan. By definition of $\tilde{\theta}$,

$$U^O(\tilde{\theta}) = U^P(\tilde{\theta}; p), \quad \text{that is,} \quad u_0 = \frac{1}{2c}\tilde{\theta}^2 - p.$$

Solving for $\tilde{\theta}$ yields $\tilde{\theta} = \sqrt{2c(u_0 + p)}$.

We now construct a cap $\hat{L}$ such that the free plan is exactly as attractive as the other two options at type $\tilde{\theta}$. Since $\theta_{FP}(p, L) = cL + \sqrt{2cp}$, the condition $\theta_{FP}(p, \hat{L}) = \tilde{\theta}$ leads to

$$\hat{L} = \frac{\sqrt{2c(u_0 + p)} - \sqrt{2cp}}{c}.$$

Because $u_0 > 0$, the constructed cap $\hat{L}$ is strictly positive. Moreover, the inequality $\hat{L} \leq \sqrt{2u_0/c}$ holds for all $p \geq 0$, since it is equivalent to $\sqrt{u_0 + p} \leq \sqrt{u_0} + \sqrt{p}$. Hence, at $L = \hat{L}$, the outside option–free plan boundary lies in the binding-cap regime, where

$$\theta_{OF}(\hat{L}) = \frac{u_0}{\hat{L}} + \frac{1}{2}c\hat{L} = \sqrt{2c(u_0 + p)}.$$

Thus, under $(p, \hat{L})$, types below $\tilde{\theta}$ weakly prefer the outside option, types above $\tilde{\theta}$ strictly prefer the premium plan, and type $\tilde{\theta}$ is indifferent among all three options. No type strictly prefers the free plan, and the induced allocation is identical to the original allocation in which only the outside option and the premium plan were chosen. Therefore, any configuration that induces only outside-option and premium users can be represented by some (p, L) satisfying $\theta_{OF}(L) = \theta_{FP}(p, L)$.

A.2 Proof of Lemma 3.1 (Binding-Cap Optimality)

We show that any optimal mechanism satisfies $L^* \leq \sqrt{2u_0/c}$, ensuring all free-plan users are constrained by the cap.

Consider any mechanism (p, L) with $L > \sqrt{2u_0/c}$, and define $\tilde{L} \equiv \sqrt{2u_0/c}$. At $\tilde{L}$, the outside option-free boundary is $\theta_{OF}(\tilde{L}) = \sqrt{2cu_0}$, and there are no unconstrained free-plan users since $c\tilde{L} = \theta_{OF}(\tilde{L})$. To preserve the free-premium boundary, choose $\tilde{p}$ such that $\theta_{FP}(\tilde{p}, \tilde{L}) = \theta_{FP}(p, L)$. Using $\theta_{FP}(p, L) = cL + \sqrt{2cp}$, this yields

$$\tilde{p} = \frac{\left(c(L - \tilde{L}) + \sqrt{2c\tilde{p}}\right)^2}{2c} > p.$$

Comparing profits between (p, L) and $(\tilde{p}, \tilde{L})$:

- (1) For types $\theta \leq \theta_{OF}(\tilde{L}) = \sqrt{2cu_0}$, profit is unchanged (outside option under both).
- (2) For types $\sqrt{2cu_0} < \theta \leq \theta_{FP}(\tilde{p}, \tilde{L}) = \theta_{FP}(p, L)$, the consumer remains on the free plan. Reducing the cap weakly decreases usage and therefore weakly reduces the firm's cost, while revenue r is unchanged. Hence, profit weakly increases.
- (3) For types $\theta > \theta_{FP}(\tilde{p}, \tilde{L})$, the consumer chooses premium under both mechanisms. Since $\tilde{p} > p$, premium revenue strictly increases while usage and cost are unaffected, so profit strictly increases.

Thus $\pi(\theta; \tilde{p}, \tilde{L}) \geq \pi(\theta; p, L)$ for all $\theta \in \Theta$. Since pointwise dominance implies LCE dominance, $(\tilde{p}, \tilde{L})$ yields weakly higher robust expected profit. No optimal mechanism induces unconstrained free-plan users.

A.3 Proof of Proposition 4.1

We show that the first, second, and fourth LCE cases (illustrated in Figure 1) cannot occur at an optimal mechanism, leaving only the knife-edge third case.

Ruling out the First Case (Figure 1a). If π_H is so low that this case applies, then a small increase in p strictly raises π_H without affecting plan choices for types below θ_{FP} , thereby strictly increasing the LCE at every θ . This contradicts optimality. Hence, an optimal mechanism is inconsistent with the first case.

Ruling out the Second Case (Figure 1b). Suppose (p, L) is an optimal mechanism and that it induces the second case. Recall that in this case, the lower convex envelope ignores θ_{FP} and is given by (note that we explicitly express the quantities as functions of p and L)

$$\pi^{\text{LCE}}(\theta; p, L) = \begin{cases} 0 & \text{for } \theta \in [\underline{\theta}, \theta_{OF}(L)], \\ \frac{\theta - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p) & \text{for } \theta \in (\theta_{OF}(L), \bar{\theta}], \end{cases}$$

and the defining inequality for this case is that the chord

$$\ell(\theta; p, L) \equiv \frac{\theta - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p)$$

lies strictly below the free-plan plateau $\pi_F(L)$ for all $\theta \in (\theta_{OF}(L), \theta_{FP}(p, L)]$. Because the chord $\ell(\theta; p, L)$ has a positive slope, this holds if and only if $\ell(\theta_{FP}(p, L); p, L) < \pi_F(L)$, that is,

$$\frac{\theta_{FP}(p, L) - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p) < \pi_F(L).$$

Fix such a pair (p, L) and consider a small increase in the premium price to $p' = p + \varepsilon$, where $\varepsilon > 0$. This leaves $\theta_{OF}(L)$ and $\pi_F(L)$ unchanged, but raises the high-type profit to $\pi_H(p + \varepsilon)$, and shifts the free-premium cutoff to $\theta_{FP}(p + \varepsilon, L) > \theta_{FP}(p, L)$. By continuity and the strict inequality at p , there exists $\bar{\varepsilon} > 0$ such that for all $\varepsilon \in (0, \bar{\varepsilon}]$,

$$\frac{\theta_{FP}(p + \varepsilon, L) - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p + \varepsilon) < \pi_F(L).$$

In particular, for all sufficiently small $\varepsilon > 0$, the mechanism (p', L) still satisfies the defining condition of this case, so the qualitative shape of the LCE is unchanged.

For such ε , the new lower convex envelope is therefore $\pi^{\text{LCE}}(\theta; p', L)$. Comparing with $\pi^{\text{LCE}}(\theta; p, L)$,

$$\pi^{\text{LCE}}(\theta; p', L) - \pi^{\text{LCE}}(\theta; p, L) = \begin{cases} 0 & \text{for } \theta \in [\underline{\theta}, \theta_{OF}(L)], \\ \frac{\theta - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \varepsilon & \text{for } \theta \in (\theta_{OF}(L), \bar{\theta}]. \end{cases}$$

Hence, for every $\theta > \theta_{OF}(L)$, the LCE strictly increases when p is raised slightly. In particular, recall that an optimal mechanism must satisfy $m > \theta_{OF}(L)$, so we have $\pi^{\text{LCE}}(m; p', L) > \pi^{\text{LCE}}(m; p, L)$. This implies that (p, L) cannot be optimal at such a mean. Therefore, the second case is inconsistent with optimality.

Ruling out the Fourth Case (Figure 1d). Suppose, toward a contradiction, that (p, L) is an optimal mechanism and that it induces the fourth case. In this case, the lower convex envelope has vertices at 0, θ_{OF} , θ_{FP} , and $\bar{\theta}$, and is given by

$$\pi^{\text{LCE}}(\theta; p, L) = \begin{cases} 0 & \text{for } \theta \in [\underline{\theta}, \theta_{OF}(L)], \\ \frac{\theta - \theta_{OF}(L)}{\theta_{FP}(p, L) - \theta_{OF}(L)} \pi_F(L) & \text{for } \theta \in (\theta_{OF}(L), \theta_{FP}(p, L)], \\ \pi_F(L) + \frac{\theta - \theta_{FP}(p, L)}{\bar{\theta} - \theta_{FP}(p, L)} (\pi_H(p) - \pi_F(L)) & \text{for } \theta \in (\theta_{FP}(p, L), \bar{\theta}], \end{cases}$$

and the defining feature of this fourth case is that the chord from $(\theta_{OF}(L), 0)$ to $(\bar{\theta}, \pi_H(p))$ lies strictly above the free-plan plateau π_F on some subinterval of $[\theta_{OF}(L), \theta_{FP}(p, L)]$. Equivalently,

$$\ell(\theta_{FP}(p, L); p, L) \equiv \frac{\theta_{FP}(p, L) - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p) > \pi_F(L).$$

Because this inequality is strict and all objects are continuous in (p, L) , there exists a neighborhood of (p, L) such that every nearby pair (p', L') still satisfies the defining inequality of the fourth case. Hence, for small perturbations of (p, L) , the qualitative shape of the LCE remains unchanged: the envelope continues to have vertices at 0, θ_{OF} , θ_{FP} , and $\bar{\theta}$.

Recall that an optimal mechanism must satisfy $\theta_{OF}(L) < m$, so that $\pi^{\text{LCE}}(m; p, L) > 0$. We distinguish two cases, depending on the position of the mean relative to the free-premium cutoff.

- (a) $m \in (\theta_{OF}(L), \theta_{FP}(p, L))$. Fix (p, L) and consider a small decrease in the premium price to $p' = p - \varepsilon$, for $\varepsilon > 0$. For ε sufficiently small, the fourth-case inequality continues to hold at (p', L) , so the LCE still has the same piecewise linear structure. Since L is unchanged, both $\theta_{OF}(L)$ and $\pi_F(L)$ remain fixed, while the free-premium cutoff becomes $\theta_{FP}(p', L) < \theta_{FP}(p, L)$ and the high-type profit become $\pi(\bar{\theta}; p', L) < \pi_H$.

Because $m < \theta_{FP}(p, L)$ and $\theta_{FP}(p', L)$ converges to $\theta_{FP}(p, L)$ as $\varepsilon \rightarrow 0$, we can choose $\varepsilon > 0$ sufficiently small so that $\theta_{OF}(L) < m < \theta_{FP}(p', L)$. Thus, the mean lies on the first nonzero

segment of the envelope both before and after the perturbation. Under (p, L) we have

$$\pi^{\text{LCE}}(m; p, L) = \frac{m - \theta_{OF}(L)}{\theta_{FP}(p, L) - \theta_{OF}(L)} \pi_F(L),$$

while under (p', L) ,

$$\pi^{\text{LCE}}(m; p', L) = \frac{m - \theta_{OF}(L)}{\theta_{FP}(p', L) - \theta_{OF}(L)} \pi_F(L).$$

Since $\theta_{FP}(p', L) < \theta_{FP}(p, L)$, the denominator in the latter expression is smaller, and hence

$$\pi^{\text{LCE}}(m; p', L) > \pi^{\text{LCE}}(m; p, L).$$

Therefore, when $m \in (\theta_{OF}(L), \theta_{FP}(p, L))$, a small reduction in p strictly increases the lower convex envelope at the mean while leaving the qualitative shape of the envelope unchanged. This contradicts the optimality of (p, L) .

- (b) $m \geq \theta_{FP}(p, L)$. We now show that the fourth case is also inconsistent with optimality when the mean lies weakly above the free-premium cutoff. Fix (p, L) in the fourth case and consider a small reduction in the cap, $L' = L - \delta$, for $\delta > 0$. This perturbation raises the plateau profit ($\pi_F(L') > \pi_F(L)$), shifts the outside option-free boundary to the right ($\theta_{OF}(L') > \theta_{OF}(L)$), and shifts the free-premium cutoff to the left ($\theta_{FP}(p, L') < \theta_{FP}(p, L)$). Moreover, with δ sufficiently small, (p, L') remains in the fourth case.

Now evaluate the LCE at the mean $m \in (\theta_{FP}(p, L), \bar{\theta}]$. Under (p, L) , the value at m lies on the upper supporting chord between $(\theta_{FP}(p, L), \pi_F(L))$ and $(\bar{\theta}, \pi_H(p))$, and is given by

$$\pi^{\text{LCE}}(m; p, L) = \lambda(m; p, L) \pi_F(L) + (1 - \lambda(m; p, L)) \pi_H(p),$$

where

$$\lambda(m; p, L) = \frac{\bar{\theta} - m}{\bar{\theta} - \theta_{FP}(p, L)} \in (0, 1].$$

As discussed above, $\theta_{FP}(p, L') < \theta_{FP}(p, L)$, so we obtain $\lambda(m; p, L') < \lambda(m; p, L)$. Next, the defining inequality of the fourth case implies

$$\frac{\theta_{FP}(p, L) - \theta_{OF}(L)}{\bar{\theta} - \theta_{OF}(L)} \pi_H(p) > \pi_F(L).$$

Because $\theta_{FP}(p, L) < \bar{\theta}$, we have

$$\frac{\bar{\theta} - \theta_{OF}(L)}{\theta_{FP}(p, L) - \theta_{OF}(L)} > 1,$$

and therefore

$$\pi_H(p) > \frac{\bar{\theta} - \theta_{OF}(L)}{\theta_{FP}(p, L) - \theta_{OF}(L)} \pi_F(L) > \pi_F(L).$$

Using $\pi_F(L') > \pi_F(L)$, $\lambda(m; p, L') < \lambda(m; p, L)$ and $\pi_H(p) > \pi_F(L)$, we can write

$$\begin{aligned} \pi^{\text{LCE}}(m; p, L') &= \lambda(m; p, L') \pi_F(L') + (1 - \lambda(m; p, L')) \pi_H(p) \\ &> \lambda(m; p, L') \pi_F(L) + (1 - \lambda(m; p, L')) \pi_H(p) \\ &> \lambda(m; p, L) \pi_F(L) + (1 - \lambda(m; p, L)) \pi_H(p) \\ &= \pi^{\text{LCE}}(m; p, L). \end{aligned}$$

Thus, a small reduction in L strictly raises the value of the lower convex envelope at the mean, while keeping the mechanism in the fourth case. This contradicts the optimality of (p, L) .

In summary, for any mechanism (p, L) that induces the fourth case, there exists a nearby perturbation of (p, L) that preserves the fourth-case geometry but strictly raises the lower convex envelope at the mean. Hence, no optimal mechanism can be consistent with the fourth case. Combined with the arguments for the first and second cases, this establishes that any optimal mechanism must satisfy the knife-edge collinearity condition (Figure 1c). $\square$